The Tungsten Market Is Warning of an Upcoming War

Podcast: Odd lots

AI Executive Summary

Core Thesis Tungsten serves as a century-scale "prediction market" for global conflict; its demand and economic viability fluctuate in direct correlation with geopolitical tensions. While China currently dominates approximately 80% of the global supply, the rise of modern warfare (drones, precision munitions) and increasing geopolitical instability are driving a resurgence in Western mining interest. However, investing in this sector remains difficult due to the lack of standardized futures markets and the extreme price volatility inherent in a small-scale, non-hedgable commodity market.

Key Arguments & Data * **Strategic Physical Properties:** Tungsten's utility is driven by its extreme density (comparable to gold) and an incredibly high melting point (exceeding 3,000°C). These properties make it essential for armor-piercing projectiles, drone shrapnel (cluster munitions), turbine blades, and tungsten carbide tooling used in heavy drilling. * **Market Dominance & Geopolitics:** China controls roughly 80% of the global tungsten supply. The use of export controls by China serves as a significant geopolitical lever, prompting Western nations to seek alternative sources to avoid "chokeholds" similar to those seen in rare earth elements or gallium. * **The "War Cycle" Phenomenon:** Historically, tungsten mining follows a boom-bust cycle tied to major conflicts. The discussed Australian mine has opened and closed in response to WWI, WWII, the Korean War, and the Vietnam War; its current reopening is driven by the tensions of the Ukraine war and potential conflict in Taiwan. able * **Economic Constraints & Risk:** Unlike copper or gold, tungsten lacks a robust futures market or forward curve. This absence of a way to hedge costs makes it nearly impossible for traditional banks to finance new mining projects, as revenue projections are highly unpredictable. * **Price Volatility Metrics:** The price of ammonium paratungstate (a tradable form of tungsten) in Rotterdam has seen massive swings, rising from approximately $300 per dry metric ton unit (dmtu) in 2022 to over $3,000 recently. * **Micro-scale Investment vs. Macro-scale Impact:** The capital required to revitalize critical mineral mines is remarkably small relative to global defense or tech spending; for example, the discussed mine required

roughly $77 million over 20 years—a sum dwarfed by the cost of a single advanced missile system or large-scale AI infrastructure. * **The Substitution & Exploration Effect:** While extreme scarcity can lead to technological substitution (e.g., moving away from cobalt in batteries), moderate scarcity encourages "wildcat" geological exploration in alternative regions like Brazil, Malaysia, and Africa.

Actionable Takeaways * Monitor Critical Mineral Supply Chains: Investors and policymakers should track the "chokehold" potential of minerals like tungsten, gallium, and germanium. The shift from "just-in-time" efficiency to "just-in-case" security is a primary driver of current capital flows. * **Watch for Government Intervention:** Look for shifts in Western trade policy, such as the implementation of "floor prices" or strategic mineral reserves (e.g., the US "Project Vault"), which are designed to provide the price security necessary to attract private mining capital. * **Assess Geopolitical Risk via Commodities:** Use tungsten and other strategic metals as leading indicators for shifts in global military posture. A move toward localized, non-Chinese production signals a long-term expectation of persistent geopolitical friction rather than a return to globalized peacetime equilibrium.

Full Transcript

Hi, I'm Barry Rithaltz, inviting you to join me for the Masters in Business podcast. Every week we bring you conversations with the people who shape markets, investing and business. I speak with CEOs, Nobel laureates, market innovators and legendary investors. Whether you own stocks, bonds, real estate commodities, even crypto, these are discussions you absolutely need to hear. Subscribe to the Masters in Business podcast on Apple Spotify or anywhere you listen. Bloomberg Audio Studios. Podcasts. Radio. News. Hello and welcome to another episode of the OddLots podcast. I'm Joe Weisenthal. And I'm Tracy Allaway. Tracy, I'm in a mood of a little bit of heartbreak this morning. Oh, mood of a little bit of heartbreak. That sounds like a country single, doesn't it? That's right.

No. I know what you're sad about. You're singing the tungsten cube blues. The tungsten blues. I'm actually a little shook. In 2022, remember the tungsten cube hype? Unfortunately. I bought one and it was the only and it feels so good like a nice big block of tongues in your hand. It feels so good. It's so unexpectedly heavy. Probably like a little like mild autism like I like related like sort of like heavy things like weighted blankets and heavy objects. And I was going to bring it to this episode because I know you love prop sometimes. I do like a good prop. And I was like, I'll bring a prop. I couldn't find it on my desk. So we did a desk move recently and either somehow I I hope I so find it either it got misplaced in the move or because the value of tungsten has gone up so much.

Maybe someone stole it. I think I know which one it is Joe. I think I know. Let me just say. I have a messy desk too. For listeners who were lucky enough to have missed out on the tungsten cube craze, there was a moment where for some reason it was mostly crypto bros on Twitter all seemed to go nuts for buying these tungsten cubes and then showing pictures of them on social media and then promptly like Joe forgot about them and actually this is a really good analogy for what we're about to talk about because we're going to talk about tungsten the metal and tungsten it turns out has gone through these cycles where for a few years it's an absolutely essential strategic asset that you need in order to declare war and then as soon as the

war is over you kind of forget about it you still need it for some industrial applications or for novelty desktops but then You find out that you need it again. And so there's this long running kind of boom bus cycle. I'm really sad. You're a demonstrative example of how everyone seems to forget about tungsten, except China. I'm actually really sad about this. This is, by the way, a little bit of... Can I say one more thing? Yeah, please, please. I know where you can find some tungsten. In Western Australia? In the Bahamas. Oh, really? FTX, famously. Oh, that's right. That's right. Yes.

was going to install a giant tungsten cube at their headquarters in the Bahamas. And everyone seems to have forgotten this bit of the Michael Lewis book at the very, very end of Michael Lewis's book on Sam Bateman Freed. He says he found the giant tungsten cube in a storage unit in the jungle. And it's probably still there because no one would want to move it. Yeah, it's so, it would be so heavy. I have to, it's actually depressing me. You know, one of the reasons like I didn't buy the cube as a quote investment unquote. But I think I've always said, you know, one of the things that people don't talk about with hard asset investing is that you actually pay a negative interest rate because A, you have to either pay for storage, constant storage, like a safe or something like that. Or if you have a storage unit in the jungle. Or eventually security guards, right? If you really had a lot of gold or silver or whatever. But then if you don't have that, if you don't do that, then every year there is like a non-zero percent chance that you lose.

your asset you misplace it. Maybe you have like some keys to a crypto wallet that you lose or whatever. So that is a way of replicating except sort of stochastically the negative interest rate. I think I'm gonna find it. I didn't totally forget about it. I had it on my desk even as recently as a few months ago and I have some very messy drawers. So I think I actually, I haven't actually given up hope. I'll create a bet on a prediction market about whether you'll find your tungsten cube. But anyway. Setting aside desk toys, it turns out tungsten is actually, as you said, important sometimes. You know, it's not like copper or oil where it's important all the times. It seems to be a commodity that's important sometimes. And that creates, as you said, interesting dynamics. Yeah. And it seems to be growing more important because going back to the military applications of tungsten.

We've had a lot of wars recently, and Tungsten apparently is in a lot of cruise missiles and drone shrapnel and things like that. And at the same time, like a lot of other, I guess, rare earth metals, it's also becoming more important to our vision of the future, to the AI build out, things like that. So it's getting more attention. And of course, again, like a lot of rare earths, there's a limited supply, and it seems to be dominated by one particular country. Yeah, this is the thing. There's a lot of it. But it's expensive at low margin in typical times. And so it's like, yeah, it's like distributed and there's some in countries that are friendly to the U.S. that are like more seen as geostrategic rivals. But in normal times, it's just not worth the squeeze to actually spend to spend the time and environmental costs of mining it. So we should talk about.

the current state of Tungsten in a state of heightened geopolitical tensions. Let's do it. Anyway, I'm very excited to say we really do have the perfect guess, someone who we probably should have had on odd lots. A long time ago, one of our colleagues coming to us from Sydney, we're going to be speaking of course to Bloomberg opinion columnist David Fickling, who has just done a big reported piece about a Tungsten mine in Australia and what this tells us about all the things that we're talking about. David, thank you so much for joining us. How are you doing? Great to see you. Thank you so much for staying up late to chat with us. Why would we talk about war, geopolitical uncertainty, all this stuff? What is it about tungsten that suddenly becomes important? Yeah, I mean, tungsten, you can look at it as a bit like a sort of century old missing prediction market for war, basically. And I didn't realize this until I started looking at tungsten. But actually, if you go back right to the turn of the 20th century, it is operated like a prediction market in the sense that people making real investments once they get a sense that war is a real possibility they put real money behind tungsten and you can see the sort of process with that if you're operating a tungsten mine then you're in touch with tungsten traders and tungsten traders are in touch with military you know material companies.

who get the first big orders when there's a real prospect of war. And this mine I visited has this extraordinary history where it first opened in 1917 during World War I. It closed at the end of the war because then the war ended, no one needed the tungsten anymore, and the European government

sold it for their tungsten stockpiles. It opens up again 1938 before World War II starts. It goes through a bit of a struggle after World War II, but then it's more or less like rescued by the Korean War. It closes down almost all together, roundabout when Sputnik has launched and people start rethinking what wars are going to be like. But then it sort of keeps going through the Vietnam War and finally closes down the Cold War and it's been closed. This mine, it's on a beach in an island just off the coast of Tasmania. It's a pretty isolated windswept spot in the roaring 40s. In 1990, it closed down. The mine flooded and it was basically been ignored for most of the last three decades and it's now finally starting up again at this very moment when war is looking more likely. So this is a story, but just to be clear, what is it about the properties of tungsten such that it is important for material? I mean, it's the same thing with the tungsten cube on your desk. It's incredibly dense. I think one of the things that attract the cryptobroids to tungsten is it's not just incredibly dense. It's about the same density as gold.

But it also has this ridiculously high melting point, more than 3,000 degrees Celsius. So three times hotter than lava, tungsten will still be solid. And that means that if you're trying to make projectiles with it, for instance, if you're trying to make an armor piercing projectile to get through the armor plating of a tank, it's going to hold its shape. Most materials are going to turn into a mushroom and dissipate their force. This is going to go straight through to the inside of the tank, so it'll be very effective. Or also a thing that's more current at the moment. say you're having a sort of cluster munition, a sort of a warhead that explodes into a sort of distributed charge so that you can take out a command post or a drone. With tungsten you can have a bunch of very tiny tungsten pellets and they will make a sort of metal rain that will smash anything in the vicinity and because it's so dense it's going to be much more effective than steel. So for all these reasons it's just very useful for military purposes and of course if war is pending people pay any price for those sorts of physical properties. Getting hit in the head by a tungsten cube would certainly be painful and damaging. Talk to us about the non-military applications because it's possible, you know, you wrote about this new mine that's restarting. Maybe, well, we should get into what the investors actually say, the reasons behind them opening up this mine. Maybe it's because they're predicting more tungsten demand because of...

additional military conflict. Maybe they just think there's going to be more demand for like, I don't know, carbide tooling or something like that. What else is tungsten used for? And then also, what does, we should also talk about how colorful some of these characters are. But what is your merry band of tungsten investors say about their reasons for opening it? You've done your research, Tracy, haven't you? I mean, this carbide tooling, I could tell that was the... I mean, the key reason you will use tungsten is, again, because it's very hard, it's basically drill bits. If you're in an auto shop, then you will use the drill bit for working metal. If you're in a mine, it will be on the end of a drill bit there or in an oil well, anywhere where you need to drill something very hard. Tungsten carbide is a mixture of tungsten and carbon, as the name suggests. And it's the...

Best value way to get incredibly incredibly sort of powerful sort of drilling power and that's actually essentially in all sorts of ways But I think we don't often anticipate I was in the course of working on this article I was reading something about during World War two when Nazi Germany when we can talk about this later was very Concerned about its supplies of tungsten and they were running short of supplies of tungsten And one of the things they did is to conserve it for the weapons industry. They stopped using it in tooling. And you hear these reports of Allied soldiers capturing German tanks late in World War Two, and just going, this equipment's terrible, because they didn't have the tungsten carbide for the tooling. So the machinery wasn't working very well. The gears were all out of whack, and nothing was working very well. And so most tungsten now, 80% of it is just tungsten carbide tools. And you can buy it in your local tool shop.

you'll find it everywhere and it's a fairly boring hum drum element. The other things there are a few other uses for it like almost every critical mineral it gets used in some weird processes in oil refineries. If you're as into critical mineral design there's always like five percent it's always used as some weird catalyst in oil refinery and that's the case here. Another one that's actually quite interesting again because of these sort of high temperature, high strength properties if you're making turbine blades it's very important for turbine blades a lot of you know in a jet turbine but also in a gas turbine which gets used a lot of powering data sensors and things in LNG compressors about up to 10 percent of that alloy might be tungsten as

well so there's a certain amount of demand for that and there is even supposedly a use in some AI chips for tungsten as well but the quantities there will be absolutely tiny because of course anything involved in chip making the raw materials is just infinitesimal because the The chips themselves are slightly small. So the investors in this mine that's reopening the dolphin mine, I called them a colorful bunch earlier and it really does, I think you make this point in your piece, but it really does sound like a cast of characters from like a heist movie. It's like a red bull heir and like some distressed debt guys and then some retired miners doing one last job. But when they talk to you about their reasons for reopening this mine, what did they say?

Well, the main one I was speaking to was actually the chief executive or the executive chairman of the mine himself. He was actually, he was brought in by one of these investors. As I said, if you look at the shareholder register of this company, it's like nothing I've ever seen of a shareholder register, certainly of a company of sort of global importance, like you say. I think the reason Red Bull were involved is that there's a sort of long central European Germanic interest in tungsten, unreadable based in Austria. The Austrian urge to acquire tungsten, I'm well aware. Yes, absolutely. The Peabody Energy, the American coal mining company bought a coal mine off a couple of Australian investors about 20 years ago, and they used the payout from that to invest in this mine, they were some of the first people who were involved.

There's a caterpillar dealer who actually worked at the mine in the 1960s. He's one of the shareholders that the state government of Tasmania is one of the shareholders because they were bailing it out. And even some of the companies that were leasing cranes and mining equipment, they basically took equity in lieu of unpaid bills. There's really hardly anything in the way of bank finance in this company at all. Its equity is about seven million Australian dollars. And the total cash some invested Since it started up, really tried to get back into operations 20 years ago. There's been a lot of reports and environmental studies and things over that period. Total is about 77 million, you know what Amazon spends in about four hours. So it's really very little money. I would say their ambition, they weren't betting for

20 years on war, but their ambition was that China produces 80% of the world's tungsten.

and by 2010 you were seeing China threatening Japan for instance over its rare earth supplies and you could see this was a potential risk even back then and the people around the world there was one tiny tungsten mine that the the US is last tungsten mine was actually just east of Los Angeles it was a sort of retired Vietnam veteran who would go up to the mountains and like dig out a few tons of tungsten every summer and that went on until he went on mining until about 2013 I think. He was talking back then in the process of writing this article I dug out a letter that he'd sent to Congress complaining about reducing tariffs on Chinese tungsten because he was concerned back in 2012 that trying to control too much of this and was strategic competitor. Back then people weren't worried about it very much. Now it's everything everyone worries about.

Gain insight on the innovators, disruptors, and tech-driven trends shaping today's complex economy. I'm Carol Masser. And I'm Tim Stenevich. Wrap up your work day with the Bloomberg Business Week Daily Podcast. We bring you deeper dives into the story shaping your world from the evolution of AI to the shifting priorities of global business. Plus Silicon Valley power players and the latest tech trends. Catch up on the conversations you miss during the day. Subscribe to the Bloomberg Business Week Daily Podcast on Apple, Spotify, or anywhere.

you listen. So I definitely want to talk more about the sort of economics and significance of this mine. But can you just talk a little bit more? You mentioned China producing 80% of the world's tungsten. Can you just give us a little bit more color on what the peacetime tungsten market looks like in terms of how big we're talking about the sort of the I guess I would say the why there hasn't been so much investment like like what are the economics such that generally during peacetime it does not make sense to do a new project given either the size of the market and maybe presumably the cost competitiveness and the efficiency of the Chinese miners.

I mean I think the biggest thing and I think this gets underestimated a bit because China does try to lock down a lot of the supply chain in a lot of critical resources but in the case of tungsten and I think it's actually similar

in the case of rare earths it has just happened to get the best geological resource of this by far and that's a thing that happens in mining South Africa and Botswana have the best geological resources of diamonds. And Russia and South Africa have the best resources for platinum group metals. And Australia and Brazil have the best iron ore. That just happens. These resources are unevenly distributed around the world. And it's hard if you don't have resources as good as that to get a look in. China, going back to the turn of the 20th century, dominated global tungsten supply. And the only time that that's really changed was during a period in the Cold War when tungsten was basically closed off from the rest of the world. As soon as Chinese Tungsten was able to get into the world market. Their mines are the highest grade mines and there is just more of it there. Now, some investors outside China and Tungsten, alongside the geopolitical argument, they would probably say, look, these mines have been mined for a century. One of the reasons China at this moment has export controls on Tungsten, I think it's probably mostly geopolitics, but the defence of that that the Chinese government might have, is that these mines are very old and we need to reorganize the industry because it's an underperforming industry. So there is a fairly scarce resource of this. It's about 85,000 metric tons a year is all the tungsten that's used. It's not a super exciting market because it's very expensive and because it's mainly used in places like tool shops and mines. Most of that stuff gets recycled if you're in an auto tooling shop.

Once your drill bit gets worn out, it might get reground or you might just send it and it'll be recycled. So there's not a huge ongoing demand surge for tungsten. It really grows at the pace of pooling capacity worldwide. Can you talk to us about the market structure of tungsten as well? Because I know Joe said he didn't buy the tungsten cube as an investment. But if you had, I mean, you would have a hard time actually looking up the price of tungsten right now because it doesn't seem to have a forward curve in the way of a traditional commodity like oil or something like that. So if you're an investor trying to make a bet on a tungsten mine, it seems really hard to predict what your current returns are going to be much less your future returns. I think that's absolutely right. And I think it's one of the biggest problems with critical minerals generally because there's not a futures market.

because there's no way of hedging your costs. With any mine, you are making capital investments right now for something that might only pay off in years five to ten. Now, if I'm doing that with Copper, I can go to the London Metal Exchange or Comix, and I can have as many contracts as I want, setting the price of Copper five years hence. And then I can take that to the bank and say, look, here's my operating costs. Here's the value of Copper. Here's how much I'm going to produce. This is a bankable project. very simple. Totally different with Tungsten. There simply is no futures market in this because it's such a tiny market. Most of the time, one investor I spoke to, I think he's been on a podcast before actually, Alex Turnbull, he said sometimes it could be like Tungsten, it could be 300, a dry metric ton unit, which is the unit they use, it could be 3,000, 300, 3,000, Each is a reasonable price for it, because depending on the state of supply and demand of the market, the fact that China may suddenly close off supply and China has 80% of supply and then you have most of the world fighting over the remaining 20%, any of these prices is viable. And that means that if you're stuck in the middle and most tungsten mines outside China are above that a ton level, then you find it very hard to finance your mind. No banker will really lend money to you on that basis because they don't know what your revenues are going to be next year, let alone five years from now. There's no forward curve at Tungsten, but we do have one ticker on the terminal, which is Tungsten held in warehouses in Rotterdam. Oh my God, this chart just makes me want to cry. It was literally, it was probably about 300 for a ton when I bought it sometime in 2022, and it is over 3,000 right now. So if I wanted to Yeah, I should say that's three thousand dollars for this thing this weird thing, which is a dry metric ton unit. It's actually about four hundred thousand dollars for a ton of tungsten Wait, what actually this is helpful great. So when I actually so when I see What that three thousand what are what is this? I like learning stuff like this. What are these prices even mean? So I had to spend like half a day trying to work out what this even means So the price you're seeing is a thing called ammonium paratunct state, which is a compound of tungsten And it's a nice, easily tradable form of tungsten. You can just put it in sacks and ship it around the world. And then I think you heat it and the ammonia comes off and you've got tungsten oxide and then you can quite easily turn it into a lot of other things. And that is priced, quoted in dry metric ton units, which a few

critical minerals are quoted in these units. A dry metric ton unit contains about, I think, 7.93 kilograms of tungsten.

Okay. So if it's 3,000 a diametric ton unit, it's about 400,000 for a ton of tungsten. So that's why someone stole your cube. It's good money. I'm really, I'm really depressed about this. I got to find it. I got to find it. Oh my God. I'm so this is this is really bothering me. All right, talk to us about, okay, this particular mine. Walk us through a little bit more of the sort of the economics or the best case economics, let's say it works out, what are we talking about in terms of how much investment? What is sort of how much it could shift in terms of, you know, Australia's ability to where would it change in terms of market share and so forth? And like, what would it represent if it were successful? The thing that fascinates me about critical minerals is that the numbers involved are often absolutely tiny. Like I said, Total cash some invested in this in 20 years is 77 million. It's nothing. I mean, I wrote a piece about critical minerals a few a few months ago. So, you know, think of the price of a White House ballroom, which I think are we talking like 800 million, a billion dollars now? I can't remember what the latest. But I mean, you could solve the entire rare earth's problem.

for the price of that. And actually some of the investors who've been putting serious money into solving the rare earth's problem with China, that's the order of money that they're spending. It's in the hundreds of millions of dollars. And this is a time when we're spending billions of dollars on AI or a single FAD missile, one of these missiles that uses a bit of tungsten, a single FAD missile costs about 10 million dollars. So The sums that we're talking about are absolutely infinitesimal. And the amount that it can move the market, of course, is a very small market. This mine I visited down in Tasmania, if it gets up and running, it should be able to supply about 2.5% of the global market. Now that doesn't sound a lot, but remember China supplies 80% of the global market. So you're actually talking 2.5% of the remaining 20%. And actually, North Korea and Russia supply another 5%. So it's 2.5% of the remaining 15%. It's quite a significant percentage. If you go back to sort of old commodities reports from the Cold War, people would talk about the free world market. So that 2.5% is roughly one sixth of the free world market. So for a very small sum, that's the sort of thing that

you can acquire in critical minerals. And yet no one is prepared to invest in it most of the time.

it's very similar to the selling down of the helium reserve. Yeah, that's what this reminds me of. Which is like, the numbers are, it's so penny wise or whatever, you know, it's so like, here's this thing that, okay, could be strategic in the future and we're gonna sell it for like half a billion or whatever. Yeah, I was thinking the same thing. Well, also just going back to China, so I take the point that, I mean, China geographically does have an advantage when it comes to a lot of rare earths, including tungsten, but it also feels like China, is more willing to tolerate lower returns for some of these industries. And I guess treat the market like a little bit more like a chessboard rather than just a sort of short-term spreadsheet. Does that mean if anyone in the West tries to start up a new tungsten mine, can China just come in and flood the market very easily and suppress prices given that they're still the dominant producer and literally anyone else is, as you just outlined, fraction of total production? Yeah, I mean that and that is one of the main reasons that investors stay well away from this. It's one of the main investors, the reasons that they've stayed well away from rare earths. At the moment, one of the reasons tungsten prices are high is precisely because China imposed export controls just around about the time President Trump came to office last year. That doesn't mean that exports stop altogether, but it means that if you're dependent on Chinese tungsten exports, which every tungsten consumer is dependent on, you want to get your supplies early before those export controls result in the Chinese government saying, no, you can't sell it to them. So as soon as those export controls are taken off, then that sort of speculative action that's going on in the market, that's one of the main things that's driving up prices at the moment, that speculative action goes away. And so that would certainly bring down prices and then in turn bankrupt a lot of miners who were dependent on those higher prices. So that's absolutely something that has happened in the past and not just the Chinese government, as I said, the first time this mine that I went to shut down in 1920 after World War I, it was because the Allies in World War I sold off all their tungsten supplies and flooded the market. A bit like, you know, the Bank of England with gold in the 1990s or any other mistaken sort of government sale of a commodity. You know, obviously anything that is geostrategic or using warfare will

have a spike in investment and spending during a war and then it will come down, obviously, but you know, we have the defense industrial complex that there is an extraordinary amount of spending on munitions and so forth that goes on in peace and wartime, et cetera, producing new missiles and there's a fortune to be made in that at any point during, again, what we would call relatively peaceful periods, maybe the 1990s or maybe like early 2010s or something like that. Were there any voices that said, look, as part of this strategic expenditure in the West, in allied countries, we should just maintain constant loss making production and investment into some of these resources because again, the numbers compared to other defense or strategic spending just do not seem very high. I think there was a real lack of that and I think one of the interesting things that I sort of realized as I was working on this piece is that part of tungsten's time in the wilderness was about a rethinking of the nature of what wars would be because in 1990 you have two things you have the end of the Cold War but then the also crucial military thing that happens is the first Gulf War and the first Gulf War causes a lot of people in the West particularly in the US to really rethink what wars are going to be like in future. Back then, people thought it would be a Vietnam type situation potentially before the war actually started that this was a million strong Saddam's army. We're going to be something that looks a little bit like Ukraine, essentially. And then, of course, it was this incredibly quick technological war. Iraq was very quickly devastated. You saw the sort of technological superiority of US military.

And that made people go, oh, OK, hang on. The Cold War is finished, so we don't need to be building up these huge anti-Soviet military arsenals. But also, we are going to be vastly technological superior to every foe that we're going to fight in war. So we need to have different types of weapons. And I can't remember. Did you have Chris Miller on the chip war? Yeah, I think he describes that really well, that dynamic, how basically chips won the Cold War. and that caused a complete rethinking of military strategy and so at that point tungsten doesn't matter so much because if you are so technologically sophisticated that you can take out enemy command posts very quickly you can paralyze their electronic infrastructure and their entire sort of military command and then you don't really need to be fighting this sort of war of attrition which is where tungsten really comes into its own now we've of course had another revolution in military affairs, we've had

another sort of rethinking of what wars are going to be like, because we're four years now into the war in Ukraine. And we've seen in Ukraine a battlefield that looks like the big battlefields of the 20th century, and it's still grinding on. And so that makes you rethink our future wars going to look like that or not. And of course, the other thing that's looming in the future is Taiwan, a conflict with Taiwan, where you're going to be up against the biggest manufacturing power on the planet, are you going to be able to technologically overmaster China in the way that the US in 1990 overmaster Iraq? I don't think people expect that to be nearly so easy. So tungsten again comes into its own. Yeah. So I'm looking at a headline right now saying the US is inviting proposals to boost supply of tungsten and 12 other critical minerals. So if you were going to encourage or make suggestions for encouraging tungsten production in the US or elsewhere. What would you be suggesting at the moment? Because there does seem to be this real tension between a potentially strategically critical resource and then I guess the limits of private capital and people actually wanting to fund something that will pay off with a real return. I mean, ultimately you need to have some sort of a floor price.

There is a huge debate that has sparked in the critical mineral space about whether floor prices are a good thing or not. A lot of people will argue that if you have a floor price for something, then it operates as a giveaway to miners and it's going to encourage sort of uneconomic supply. But I think in the context of what we're talking about here, that's actually the point. The only economic supply of tungsten is in China. So if that is a problem for you, you actually need a structure that's going to encourage the supply outside China, and it's going to encourage it for a set period. Now, there is a huge philosophical problem with this, because you can't do that with every single material that could be deemed critical. I saw an article the other day talking about fears that China was going to lock down a supply of citric acid.

Metallurgical coal has been put on the US critical minerals list. Copper is on the US critical minerals list. You can justify almost anything on this basis. And that's a real problem because the US, for instance, has set up this project vault, this 12 billion plan for some sort of a reserve of critical minerals, something similar to the strategic petroleum reserve. But it's not

very clear what the criteria will be for inclusion in project vault. And I think the risk is that project vault becomes a way of hedging price risk for fairly common commodities like copper, like nickel, like aluminium, like metallurgical coal. If the government is not very forceful, you don't actually end up getting the minerals you want in that. You just can end up with the minerals where industry wants government-backed price security. What you actually want is fairly small group of minerals where there's a real strategic issue. I think tungsten's one, I think rare earths are another things like gallium, germanium. These are some of the crucial ones. It'll be much cheaper, but you need to offer a medium term price security long enough that miners can actually borrow against that price security. So three to five years. Hi, I'm David Weston. Join me every week for the Wall Street Week podcast to hear stories of capitalism from around the world.

from geopolitical tensions and central bank decisions to artificial intelligence, energy, and infrastructure. We sit down with the CEOs, economists, policymakers, and thought leaders whose decisions are shaping markets everywhere we find them. Subscribe to the Wall Street Week podcast on Apple, Spotify, or anywhere you listen. applications. Secure your sour flavored fruit candies now. Yeah, it has some pharmaceutical applications. Citric acid excels it. It's a good tenderizer, I know that. Excels it chelating metal ions and removing mineral deposits. Scale and corrosion from industrial equipment. I did not know that. Yeah, I didn't know that either. So if there's a citric acid expert out there, maybe that'll be an episode we want to do. You know, David, there was a piece you wrote a very long time ago that has stuck with my head, maybe like coming on 10 years now, I think. And I think it was in the context of some of the fantasies that tech people have had about asteroid mining. And I believe you made the point that Okay, yes, there may be asteroids out there that have extraordinary deposits of various minerals and metals of all sorts, but we also have a place here on earth that also has extraordinary deposits of minerals and metals, and that is your country of Australia. And there's a lot of stuff just sitting out there, but it's, and it's a lot closer than space.

But it's still too far to be worthwhile to mine, which I thought was really interesting. It's like it's right there. And yet it's just there's the infrastructure doesn't exist. It's just not worth going out there. It's too much of a trek to get

out to those parts of Australia. Can you talk a little bit broader about like this sort of OK, we can say this is in the ground. But like we need we want this. This is industrially used for it's in the ground. We know where it is. We know it's rich, but it's just not worth. making the long 12-hour flight or whatever it is to get out there. I know it's more than a flight, but yeah, to get out there. Yeah, I think that the world of mineral exploration is full of this. And here in Australia, there's plenty of very small-cap mining companies, which are essentially like some of them are almost like a holiday program for a geologist. It funds the geologists.

expedition around the world with his hammers and his reagents, because some of them are really just wild exploration programs hoping to find something. We see this with tungsten. This mine is an unusual one that had been operating for about a century already, so there's an established resource there. But to give another example, a crucial group of rare earths depends on these things called ionic absorption clays, which when you get a very heavily weathered granite mountain, you end up with a lot of these clays which will capture some of the rare earths that we're most concerned about things like this probium and terbium and if you go back about five years people would say these can only be found in southern china and one of the problems in the world is that there is no way that you'll ever find ionic absorption clays anywhere except southern china but of course now we're in a world where China is putting controls on these things. It is stopping exports of a lot of these particular elements. And it's at the heart of the sort of rarer, sort of magnet supply chain concern. And believe it or not, all these ionic absorption clays are starting to turn up in other places. They're starting to find them in Brazil. They're starting to find them in Malaysia. People are starting to find them here in Australia and across parts of tropical Africa as well.

People are suddenly realising, OK, there are deposits there. They might not be as good as the Chinese deposits. There's a reason the Chinese deposits were the first to be explored. But if China becomes an unreliable resource, people are going to find them somewhere else. And I think that's, I mean, there's two twin activities that happen in commodities and mining. One is the sort of substitution effect. Sometimes you're just going to say, we can't use this. It's become too scarce. We need an alternative. I mean, a good

example of that, I think, is if you look at electric vehicle batteries, go back five, seven, eight years, people are talking about the importance of having batteries depending on cobalt. Cobalt's very heavily exposed to one country, Democratic Republic of Congo. There's a lot of child labour in the mining industry there. And also it's in some ways not quite as good. So a lot of the Chinese battery producers started using a different chemistry that doesn't use cobalt, lithium, iron, phosphate. And so So you've substituted away from that product and you don't need it anymore. And to go back to the analogy with asteroid mining, if a commodity is so scarce that you need to go to space to get it, you're probably going to substitute it for something else instead. However, if it's not quite as scarce as that, you're probably going to be able to send out geologists around the world and they're going to find it somewhere else. And if the established supply is insecure enough, then the alternative location is going to get more and more attractive.

Yeah, I think we should just substitute our tungsten use for gold. I mean, it's like the same. I was going to say, if it's the same density, you could have just bought a gold bar instead of a tungsten. You just start having gold based munitions. Those would slice right through that tank as well. David Fickling is great chatting with you. I'm so glad we made this happen. This is such a fascinating journey and great reporting that you did. So thank you so much for coming out on us. Thanks very much. It does seem so crazy to me that we have this huge, the military industrial complex, et cetera. And then there's these few things that are rounding errors. Right, the capital outlay for a tungsten mine seems pretty small in the grand scheme of things. And yet no one was like, let's just, yeah, sure, it's uneconomical, but so is all defense spending because it's loss making because there's no market use for a lot of this stuff. But it's like...

I don't know. It seems like you just allocate a few billion here and there and you could alleviate some of these potential stress points. Instead, you sell down the helium reserve over 20 years for half a billion dollars or whatever. I mean, to some extent, it does seem like a failure to predict the future, which is difficult, right? People thought that war was going to evolve in different ways. We didn't think that we would be shooting this many cruise missiles or that we would be shooting shrapnel at drones or having drones shooting this much shrapnel. And so like here we are with, to some extent,

an unexpected need for tungsten and the same for helium in some sense because we didn't know that we were going to have the semiconductor boom and things like that. So hindsight is 2020, but I don't know, it does seem like maybe we should restart that tungsten mine in California. Should we go check it out when we're...

We're in LA. Yeah, we're gonna be in a listeners. We're gonna be in LA in early September. Maybe we can find the guy who used to run it. It can go fine. Yeah, let's go check out the I also, you know, I just think that again, one of the things I always think about with respect to China specifically is the fact that there's not really a business cycle. there. And they've never had, they don't have these like, they have secular busts, like obviously real estate, etc. But you know, there's been this sort of like refusal to accept the premise that overall recessions are an inevitability. And therefore, you don't get these like periods where it's like big industries that were once important hollow out. And then it's like, Oh, it's so costly to restart them. I mean, you do get some of that. And China's pig pork industry is a pretty good example of that where there was a shortage and prices went up and so the government provided a bunch of support it over expanded prices collapsed and then a bunch of private operators went out of business so there is some of that but I do think in general China is more willing to accept just lower returns in general for some of these industries for some of these long term which also begs the question if One of the big themes of our current period in time, as we've discussed on the podcast repeatedly now, is that everyone seems to be trying to build up their own supplies, their own responses to certain chokeholds in the global economy. And so one of the concerns that you see now, and David talked about it is, well, if everyone's trying to create their own supply of tungsten, even if it's a relatively small supply, even if it doesn't actually take that much capital to create a new tungsten mine. If everyone's doing it all at once and then you don't have a definitive buyer of last resort at the end, then people are going to think that we're going to become oversupplied. But in the meantime, all of this expenditure is at the margins inflationary. And so you have the double whammy of more spending on nonproductive things because they're strategic and the diminishment because of anxiety about.

security of free trade. So you get the double lots and lots of spending and replication of spending across multiple geographies and countries. The one other thing I would say is whenever we do these commodities episodes and especially lately given what happened with oil supply in Iran, I kind of, I always marvel at the markets, I guess, creativity or ingenuity when it comes to responding to some of these shocks. So Japan had its supply of a bunch of rare earths, including tungsten cutoff. by China, which I'm sure at the time was thought to be an absolutely dire thing for a lot of Japanese industry, including car makers. But in the end, it ended up dealing with it reasonably well by sourcing recycled supply and scrap tungsten and things like that. Now, again, if China stopped exporting tungsten to everyone in the world, would there be enough scrap tungsten to replace that supply? Probably not. But for some of these supply shops, it does seem like you can get pretty good response. As we talked about with Javier Blas last year, we might have to go into a bunch of vacuum cleaners and pull out some of the magnets and stuff. Why don't you get into the scrap tungsten business, Joe? I can get you started with some metal tool bits. I had one that split the other day, which was really weird and unusual and scary. I'm done investing in tungsten, especially if I can't find this thing. I don't want to ever think about it again. All right.

No more tungsten. That's the opposite message of this entire podcast episode. Yeah, that's right. Keep thinking about tungsten. Keep thinking about tungsten. All right, shall we leave it there? Let's leave it there. This has been another episode of the OddLots podcast. I'm Tracy Allaway. You can follow me at Tracy Allaway. And I'm Joe Weisenthal. You can follow me at the stalwart. Follow our producers, Kerman Rodriguez at Kerman, Armin, Dashel Bennett at Dashbot, Kale Brooks at Kale Brooks, and Kevin Lozano at Kevin Lloyd Lozano. And for more OddLots content, go to Bloomberg.com slash OddLots for a daily newsletter and all of our episodes. And you can chat about all of these topics 24-7 in our Discord, discord.gg slash OddLots.

And if you enjoy OddLots, if you want to support a speculative OddLots mission to the Bahamas to find that giant FTX tungsten cube, then please leave us a positive review on your favorite podcast platform. And remember, if you are a Bloomberg subscriber, you can listen to all of our

episodes absolutely ad-free. All you need to do is find the Bloomberg channel on Apple Podcasts and follow the instructions there. Thanks for listening! Hi, I'm Tom Keene, inviting you to join me for the Bloomberg Surveillance Podcast. It's about making you smarter every business day. I'm Paul Sweeney. We bring you complete coverage of stocks, bonds, commodities, even crypto, all the information you need to excel in the markets. And I'm Alexis Christophorus. Listen to us for essential conversations with the smartest names in economics, finance, investment, and international relations. That's the Bloomberg Surveillance Podcast. Subscribe today on Apple, Spotify, or anywhere you listen.